

Problem 2. Avocados

- a. In 1980, less than 1% of avocados in the US were imported. California supplied the vast majority of avocados grown in the US. The supply of avocados (in millions of pounds) from California is given by:

$$Q_S = \frac{1}{2}(250P - 100)$$

Demand for avocados (in millions of pounds) in the US is given by:

$$Q_D = 400 - 100P$$

If the US only has access to California avocados, what would the equilibrium price (for a pound of avocados) and quantity be?

Price: _____

Quantity: _____

Answer. Equating supply and demand yields a price of \$2 and quantity of 200 million pounds.

- b. By 2019, avocado consumption had increased and most avocados consumed in the US were imported.¹ Mexico is by far the largest producer of avocados in the world. Given the favorable climate, Mexico is able to supply avocados at a price of \$1 per pound (and there is effectively no limit on how many can be grown).

If there is free trade with Mexico, how many millions of pounds of avocados will be consumed in the US, and what share of them will be produced in the US? Are US producers and consumers better or worse off from trade?

Quantity consumed: _____

US-produced Share: _____

- ☐ US consumers and US producers are better off
- ☐ US consumers better, US producers worse off
- ☐ US consumers worse, US producers better off
- ☐ US consumers and US producers worse off

Answer. If Mexico can supply any amount at \$1, then the market price will be \$1 (since it is less than the price found in a), and demand at $P = 1$ is $Q_D = 300$. At that price, California

¹ The true figure is approximately 90%.

farmers would only be willing to supply $Q_S = 75$, which is 25% of demand. Therefore, the US share of sales is 25%. Since the price has effectively fallen, US consumers are better off, while US producers are worse off.

- c. The government is considering imposing a tariff on imported avocados to help US farmers. Suppose they were to levy a tariff of \$0.20 per pound on imports. By how much would US consumer surplus increase or decrease from this tariff? By how much would US producer surplus increase or decrease from this tariff? How much government revenue would the tariff generate? (Hint: draw a diagram of the situation and compute areas corresponding to the different objects at the free-trade equilibrium and tariff equilibrium).

Change in US consumer surplus from tariff: _____

Change in US producer surplus from tariff: _____

Government revenue from tariff: _____

Answer. Price will increase to \$1.20, as this is still less than the no-trade price. The new equilibrium has $Q_D = 280$, $Q_S = 100$. Consumer surplus goes from $(0.5)(3)(300) = 450$ to $(0.5)(2.8)(280) = 392$, for a change of -58 . Producer surplus goes from $(0.5)(0.6)(75) = 22.5$ to $(0.5)(0.8)(100) = 40$, for a change of $+17.5$. Tariff revenue would be $\$0.2$ times the quantity imported (which is quantity demanded less domestic supply), or $(180)(0.2) = 36$.

- d. Avocado lovers are not happy about the plan and decide to lobby against it. The government proposes to hand back 100% of the tariff revenue it collects to avocado consumers at the end of the year.² Would this make avocado lovers just as well-off as under free trade? Why?

☐ Yes

☐ No

☐ Need more information

Why? _____

² Ignore the practical challenges of doing this for now; assume they have a way to do this.

Answer. No, it would not, as the tariff revenue (36) is less than the change in consumer surplus (-58). This is the key insight: even total surplus falls from the tariff.